

Deployment of Explicit Model Predictive Controller on a Mini Fuel Cell Vehicle

Amit M. Manthanwar^a, Jubeda N. Khatun^a and Efstratios N. Pistikopoulos^{a,b}

^a*Centre for Process Systems Engineering, Imperial College London, UK*

^b*Artie McFerrin Department of Chemical Engineering, Texas A&M University, USA*

Abstract

The aim of this paper is to demonstrate hardware-in-the-loop implementation of explicit model predictive controller on an embedded microcontroller platform deployed on-board a prototype fuel cell powered vehicle. The fuel cell technology holds a key to securing sustainable societal benefits by decarbonizing transport sector and stimulating the global economy. Unfortunately, cost, performance and durability still remain the barriers to the rapid commercialization of fuel cells. This paper presents a two pronged agenda of our research objectives. First research objective is under the theme “use of microcontrollers in chemical engineering” by extending the scope of optimisation and control algorithms to be ultimately deployed on actual automation hardware. Second major objective is to make further contributions to the fuel cell technology by developing power electronics platform as hardware and optimization based powertrain control as software tools. This toolchain and the proposed framework can be easily adapted for all the renewable energy system applications. The results conclude that the explicit MPC strategy is a natural choice to manage the fast and complex dynamics of fuel cell system while our proposed automation framework provides uninterruptable power to the vehicle powertrain when the vehicle is operated under uncertain driving conditions.

Keywords: Model predictive control, multi-parametric optimisation, fuel cell vehicle

1. Introduction

Technological strides in the area of optimisation and control theoretic have paved the way for the development of novel optimisation based constrained optimal control algorithms. This technological breakthrough under the umbrella of advanced process control (APC) answers the much sought after control objective question, how to attain desired profit margins while satisfying all the conditions of feasibility, stability, safety, operability and performance of a plant operating under the influence of uncertainty. Model Predictive Control (MPC) is one such advanced process control strategy where mathematical model of the process is used to predict the desired transient behaviour of the plant by computing optimal policy for current and all future control input sequences. Although the control input sequences are computed for entire prediction horizon, only the current or first control action is administered. In subsequent time steps, this procedure is repeated. Due to this receding nature, MPC strategy is also referred as a receding or moving horizon control, Kwon et al. (1982). In a review paper Mayne et al. (2000) have presented the historical developments and theoretical aspects of MPC.

Search for optimal control policies using MPC algorithm requires significant online computations. This task can be alleviated by converting online optimisation problem to an offline multi-

parametric optimisation program, Pistikopoulos et al. (2000). Strength of multi-parametric optimisation is that it computes in a given state space the entire map of all conceivable optimal control policies. These computations are performed offline which generates a map of control moves as a function of system states. These resulting piecewise affine control policies can then be easily administered online by means of simple function evaluations. In principle, MPC strategy computed efficiently via multi-parametric programming can be deployed on a cost effective microcontroller platform without necessarily requiring expensive online computational hardware, Pistikopoulos (2012). The objective of this paper is to outline step by step procedure for the hardware-in-the-loop implementation of ‘MPC-on-a-chip’ technology. For this purpose we have chosen to deploy our control algorithm on a fuel cell powered vehicle, Manthanwar and Pistikopoulos (2015).

1.1. Fuel Cell Vehicle

All electric propulsion system for low carbon transport is rapidly advancing with fuel cell vehicles being the frontrunner in forklifts and light-duty vehicles. All major automakers have already announced plans to launch commercial fuel cell vehicles. Although the industry is poised to grow exponentially, the major challenges to reach the sales volume are cost, performance and operational lifetime of the fuel cell propulsion system. A recently carried out long term real world study by Wipke et al. (2012) concluded that the majority of fuel cell system failures and forced outages ($\sim 90\%$) are due to lack of system integration. Automation and control plays a critical role in system integration. It is ultimately responsible for maintaining uninterruptable power demand under load variations, Soroush and Chmielewski (2013). The real time control of fuel cell vehicle using a well established technology of model predictive control, requires an accurate mathematical model of the system. The dynamic phenomenon taking place inside fuel cell involves complex interaction of mass transport, energy transport and electrochemical kinetics. This complexity further increases when the fuel cell dynamics are coupled with the vehicle transmission dynamics, Manthanwar and Pistikopoulos (2015). Together these uncertain dynamics determine the overall operational performance of the vehicle. In addition, the real time control requires computing machinery to run the optimisation algorithms thus adding burden to already burgeoning manufacturing costs. In order to resolve these issues, the two main goals of this paper are (a) develop a prototype fuel cell vehicle as a test platform to carry out system integration studies; and (b) further adapt multi-parametric optimisation algorithms to achieve efficient operation of the vehicle.

In next section, we present the architecture chosen for the fuel cell vehicle. In this section to follow, we present our generic framework for the deployment of explicit model predictive controllers. In the subsequent sections, we present experimental aspects including the hardware development and testing of controllers. Finally, we conclude the paper by presenting benchmark test results on a few realistic drive scenarios for controlling the speed of our vehicle.

2. Fuel Cell Vehicle Architecture

The design of Internal combustion engine has been perfected over 150 years and there are many ways to integrate hybrid architecture into automotive powertrain, Pollet et al. (2012). Most popular architectures are parallel and series hybrid with a fuel cell as the primary energy source and a battery or super-capacitor as the supplementary energy source. This arrangement is advantageous as it overcomes the problem of initial cold start-up of the vehicle or situations when demand exceeds power produced by the fuel cell. For simplicity and ease of demonstration we have chosen architecture shown in Figure 1. In this architecture, fuel cell is used to charge the battery and battery powers the vehicle. Embedded controller is used to monitor and control the speed of the vehicle for variety of operational scenarios. Our deployment goal is to control the speed of a fuel cell powertrain and demonstrate the applicability of MPC on a low cost microcontroller platform. Further goal is to outline the hardware development aspects for testing and validation of explicit

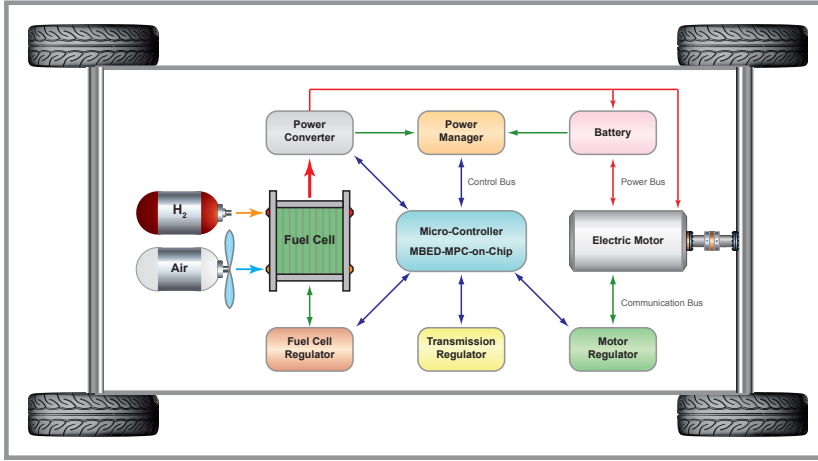


Figure 1: Architecture for Fuel Cell vehicle empowered by MPC-on-Chip technology

MPC algorithm. In the next section we present the framework for control followed by hardware for our prototype fuel cell vehicle with step by step procedure for the implementation of the proposed framework deployed on a fast “MBED” platform based on a high performance CortexTM M3 microcontroller of ARM incorporation, Yiu (2010).

3. Generic Control Framework

Major pursuit for an ideal model based control strategy is to first develop a mathematical model of the system for which controller is designed that is able to handle all process constraints. Next, implement first control sequence, and re-evaluates controller performance to further improve control strategy based on the plant’s response. On these guidelines, our generic control framework for the fuel cell powered system is shown in Figure 2. We first identify a model of our vehicle powertrain that captures the complex dynamics. This model is analysed and its accuracy is validated by estimating model parameters against the experimental observations. Using this validated model we design explicit/multi-parametric model predictive controller for the integrated system. Finally, we deploy the controller on microcontroller platform on board mini vehicle to attenuate unmeasured disturbances and filters out noise. A detailed step by step flow of our framework is shown in Figure 3.

Note that real time optimisation can be easily added to this framework to further improve the performance of regulatory control loops. This can be achieved by using a single layer or two layered approach. Thus, our framework allows for extension to any new technologies in control system hierarchy that integrates optimisation objective without compromising on system performance. In this work, we focus only on the modelling and control implementation and do not take into account objectives of real time optimisation that can potentially deliver greater value in terms of operational performance and maximise dynamic economics of process control. This approach was first proposed by Loeblein and Perkins (1999) and later improved by Peng et al. (2005) as two layered approach. Recently, integration of such strategy into a single layer is called economic MPC, Amrit et al. (2013). Additionally, our generic framework is not limited to linear controllers only. It also includes robust MPC Manthanwar et al. (2005b), Hybrid MPC Manthanwar et al. (2005a) or Nonlinear MPC Ziogou et al. (2013) strategies. The controller can be fine tuned using the tuning framework of Chmielewski and Manthanwar (2004).

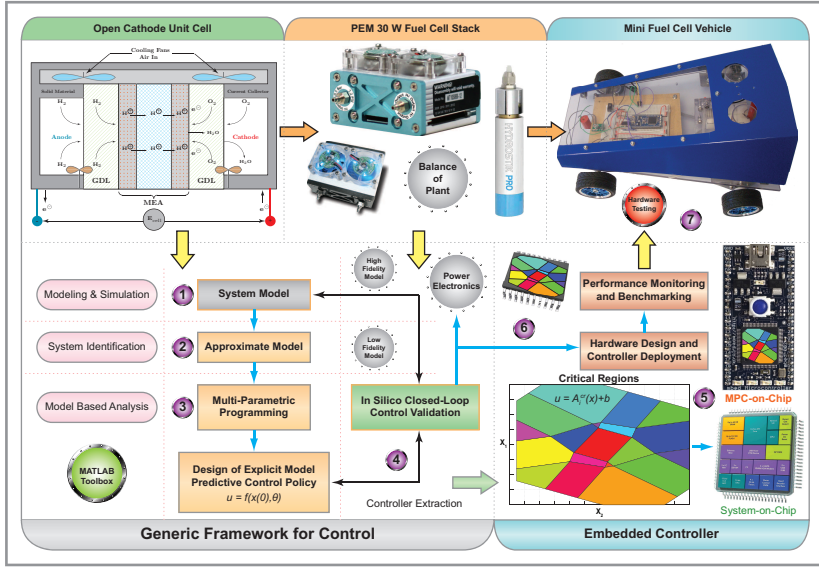


Figure 2: Generic Control Framework

4. Hardware Development

The construction of our mini fuel cell battery powered vehicle is achieved in two parts—mechanical assembly and electrical circuitry is shown in Figure 4. The power of the system is provided by a double source of energy: a fuel cell and a rechargeable battery. The fuel cell is the main source of power with a Lithium-Ion/Lithium-Polymer battery as the secondary source. We used a 30 W polymer electrolyte membrane fuel cell from Horizon Fuel Cell Technologies and two standard DC motors with built-in encoders. The chassis of our two-wheel drive vehicle was assembled with all the components (motors, wheels, electric circuit and fuel cell) to the base. The materials used to construct the mini-vehicle were 5 mm perspex sheet, 3 mm acrylic sheet, 1 mm aluminium sheet, mild steel, metal rods and four 60 mm wheels. The hardware was assembled together using screws and nuts. The vehicle was constructed and was found to be able to move forwards and backwards easily. To move left or right, additional weights were added to the front wheel areas to ensure that wheels touch the ground allowing vehicle to turn easily.

The two main components used for the construction of electrical circuit were rapid prototyping microcontroller called “MBED” from ARM based on their Cortex™ M3 Core (NXP LPC1768 with 32-bit core running at 96 MHz, 512 KB FLASH, 32 KB RAM) and the H-bridge motor controller integrated circuit (L239DNE) from Texas Instruments. The mbed microcontroller is used to input the control information and obtain readings while the H-bridge allows the adjustment of voltage input to the DC-motors. The microcontroller was programmed using C/C++ via an online compiler from mbed website (www.mbed.org) which allows user to focus on coding on any computer without any software installation. Other components used were a 12 VDC power supply, 5 V voltage regulator IC, LCD screen to display information, electrical wires, resistors, and a breadboard to mount the circuitry shown in Figure 4. The three main methods to display information from mbed are via a terminal program, by a local file system and through wireless communication (Xbee device).

The fuel cell block is monitored by the microcontroller through the use of a current sensor, voltage sensor and temperature sensors. The battery is monitored by the microcontroller through the use of the battery management system, which includes a DC charger and battery health monitoring system. A DC/DC converter is used to condition the low-voltage fuel cell power source in to

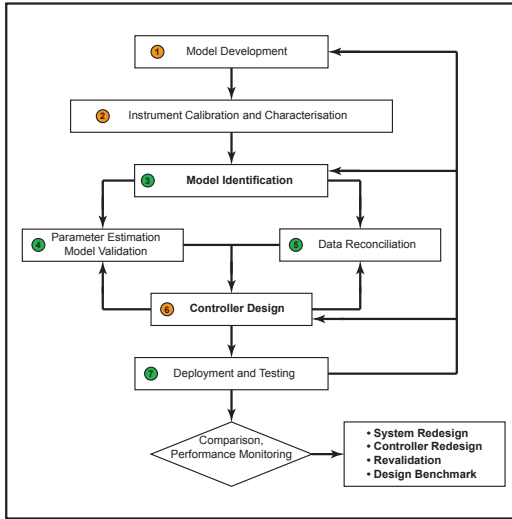


Figure 3: Deployment Flow Diagram

a constant 12V supply to the H-Bridge that drives motors and other auxiliary power demand of electronic circuit. The boosted power is also fed to charge the battery when fuel cell provides any excess power. The microcontroller monitors the DC/DC converter through the use of current and voltage sensors.

5. Results

Using the generic architecture in Figure 1 and the deployment procedure outlined in Figure 3, we first built the dynamic model of the fuel cell and vehicle transmission system, Manthanwar and Pistikopoulos (2015). Then we calibrate and characterise all the instruments including the in-house built sensor monitoring circuits shown in Figure 4. Then step change in duty cycle of the motors is given via PWM to observe the dynamic change in speed. Note that depending on the weight and surface on which vehicle is driven the speed would get affected. Our identified model considers ideal surface and we expect that the controller to eliminate any internal or external disturbances to maintain the desired speed. Figure 5 presents the result of model identification experiment.

$$A = \begin{bmatrix} 0.9635 & 4.1452 \\ -0.0029 & 0.6971 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0.0062 \end{bmatrix} u \quad (1)$$

where x is a vector of angular speed of the motor and current drawn. u is the voltage supplied. The bounds on are, angular speed $[-2 \ 2]$, current $[-0.15 \ 0.15]$ and voltage $[-1 \ 1]$. The weights of MPC objective function are $Q = [2, 0; 01]$ and $R = 1$.

6. Discussion and Conclusion

We built a prototype vehicle using two DC motor powertrain and a polymer electrolyte membrane fuel cell. The hardware development of our mini fuel cell vehicle consists of a 30W fuel cell stack and a rechargeable Lithium-Polymer battery as its primary and secondary energy sources respectively. A DC/DC converter is used to convert fuel cell generated power into a 12 volt power supply for driving motor as well as for charging the battery. The hardware comprises of a motor

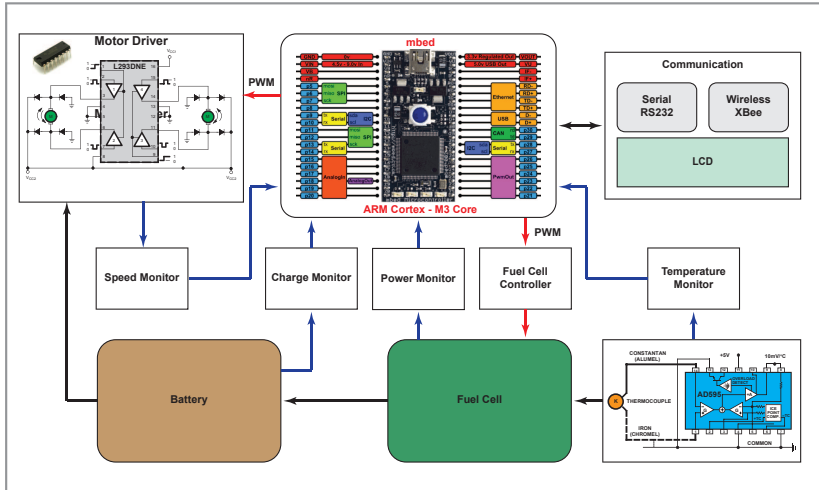


Figure 4: Embedded Control Circuitry

driver circuit as well as various sensors which are monitored by the mbed microcontroller. The sensors are well calibrated to ensure that the acquired data was reliable and produced fairly stable system noise. With the electronics produced to a high standard, step tests were carried out by performing rigorous experimentation. Obtained data was used for the model parameter identification. The parameters were obtained using the least square parameter estimation technique. The estimated parameters were used to simulate the system and matched with the experimental data. The parameters identified prove a fairly accurate model of the system with acceptable error squared value. This made it a perfect model platform to design controller. Next, we built power electronics and further signal monitoring circuits. Finally we deployed the multi-parametric controller on a microcontroller.

In conclusion, we demonstrate that the proposed embedded MPC-on-a-chip architecture is an ideal choice for the fuel cell powered systems. It helps achieve overall operational performance for the integrated fuel cell system with powertrain dynamics. The proposed architecture efficiently controls vehicle speed under the uncertain operational scenarios. The framework has an ability to easily interchange control and software functions between the different vehicle platforms and derivatives as well as different types of fuel cells. Our future goal is to test the performance of the vehicle under three uncertain scenarios namely city driving, rough terrain driving and motorway driving. A possible hardware modification to vehicle would be to make a four wheel drive to allow direct control of the front wheels.

7. Acknowledgement

The financial support from EPSRC grants (EP/I014640/1 and EP/K503381/1), the European Research Council (MOBILE, ERC Advanced Grant, No: 226462) and sponsorships from ARM Inc. is gratefully acknowledged. Authors would like to thank Chemical Engineering Workshop for their help in hardware construction of the vehicle.

References

- Amrit, R., Rawlings, J. B., Biegler, L. T., 2013. Optimizing process economics online using model predictive control. *Computers & Chemical Engineering* 58 (0), 334–343.
- Chmielewski, D. J., Manthanwar, A. M., November 2004. On the tuning of predictive controllers: Inverse optimality and the minimum variance covariance constrained control problem. *Industrial and Engineering Chemistry Research* 43 (24), 7807–7814.

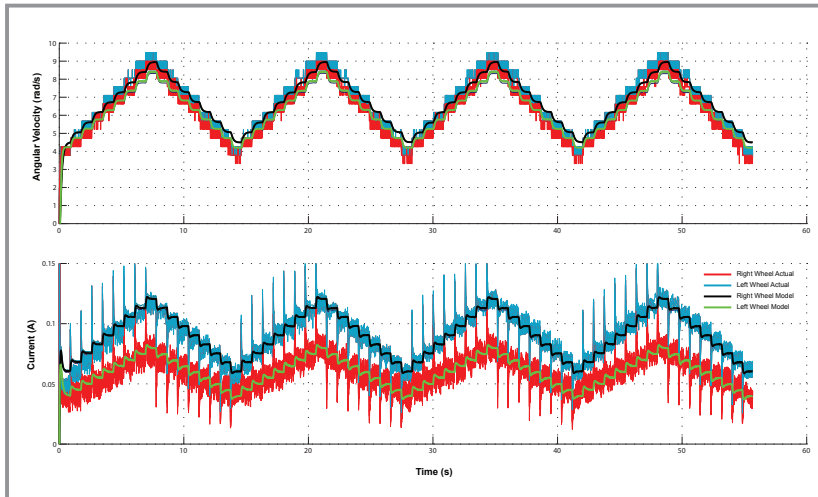


Figure 5: Model Identification Experiment

- Kwon, W., Bruckstein, A., Kailath, T., 1982. Stabilizing state-feedback design via the moving horizon method. In: Decision and Control, 1982 21st IEEE Conference on. pp. 234–239.
- Loeblein, C., Perkins, J. D., 1999. Structural design for on-line process optimization: I. Dynamic economics of MPC. *AIChE Journal* 45 (5), 1018.
- Manthanwar, A. M., Pistikopoulos, E. N., 2015. Dynamic modelling and control of fuel cell vehicle propulsion system for efficient power and heat management. *Applied Energy Special issues on Clean Transport and Energy Storage*.
- Manthanwar, A. M., Sakizlis, V., Dua, V., Pistikopoulos, E. N., 2005a. Robust model-based predictive controller for hybrid system via parametric programming. *Computer Aided Chemical Engineering* 20(C), 1249–1254.
- Manthanwar, A. M., Sakizlis, V., Pistikopoulos, E. N., 2005b. Robust parametric predictive control design for polytopically uncertain systems. In: American Control Conference. Vol. 6. pp. 3994–3999.
- Mayne, D., Rawlings, J., Rao, C., Scokaert, P., Jun. 2000. Constrained model predictive control: Stability and optimality. *Automatica* 36 (6), 789–814.
- Peng, J.-K., Manthanwar, A. M., Chmielewski, D. J., Sep. 2005. On the tuning of predictive controllers: The minimum back-off operating point selection problem. *Ind. Eng. Chem. Res.* 44 (20), 7814–7822.
- Pistikopoulos, E. N., Dec. 2012. From multi-parametric programming theory to mpc-on-a-chip multi-scale systems applications. *Computers & Chemical Engineering* 47 (0), 57–66.
- Pistikopoulos, E. N., Dua, V., Bozinis, N. A., Bemporad, A., Morari, M., 2000. On-line optimization via off-line parametric optimization tools. *Computers and Chemical Engineering* 24, 183–188.
- Pollet, B. G., Staffell, I., Shang, J. L., Dec. 2012. Current status of hybrid, battery and fuel cell electric vehicles: From electrochemistry to market prospects. *Electrochimica Acta* 84 (0), 235–249.
- Soroush, M., Chmielewski, D. J., Apr. 2013. Process systems opportunities in power generation, storage and distribution. *Computers & Chemical Engineering* 51 (0), 86–95.
- Wipke, K., Sprik, S., Kurtz, J., Ramsden, T., Ainscough, C., Saur, G., May 6–9 2012. Final results from U.S. FCEV learning demonstration. In: EVS26 International Battery, Hybrid and Fuel Cell Electric Vehicle Symposium. Los Angeles, California.
- Yiu, J., 2010. The definitive guide to the ARM[®] Cortex-M3. Elsevier.
- Ziougou, C., Pistikopoulos, E. N., Georgiadis, M. C., Voutetakis, S., Papadopolou, S., Mar. 2013. Empowering the performance of advanced nmPC by multiparametric programming—an application to a pem fuel cell system. *Ind. Eng. Chem. Res.* 52 (13), 4863–4873.